METRIC

A-A-54752
21 September 1992

COMMERCIAL ITEM DESCRIPTION

DISK, FILTERING, MICROPOROUS

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-D-37005A which is cancelled.

This Commercial Item Description covers a hydrosol type, microporous membrane filter disk suitable for quantitative surface retention of bacteria suspended in liquids.

Salient characteristics:

Shall be a white, microporous, autoclavable, membrane filter disk of the hydrosol assay type, suitable for quantitative surface retention of bacteria suspended in liquids. Filter disk shall be fabricated of cellulose ester which has a nominal diameter of 47 millimeters (mm), nominal pore size of 0.45 micrometers (um) and a thickness of 150 ± 20 um.

A grid of squares shall be imprinted on the surface of the filter disks. Each side of each square shall be 3.1 ± 0.1 mm long. The length of a side shall be the distance between two intersecting sides plus the width of one intersecting line. Grids shall extend throughout the surface, and grid marking shall be legible and permanent.

At least 77% of the filter disk shall be pores and the refractive index of the disk shall be 1.51 ± 0.01 when tested herein.

The size of the filter pads shall be such that they absorb nutrients and so that they are suitable for use with filter disk. Diameter shall be 47 mm (nominal) and thickness shall be 0.8 mm (nominal). The filter pads shall absorb 1.5 ±0.2 milliliters (ml) when tested herein.

AMSC N/A FSC 6640

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The bubble point pressure shall be a minimum of 172 kilopascals (kPa) and the flowrate shall be at least 55 ml/minute/sq. cm when tested herein. Water shall diffuse uniformly through the filter disk in 15 seconds, and there shall be no dry spots when tested herein. Serratia marceasan shall be retained (organism retention) when tested herein. The filter disk shall not inhibit bacterial growth (toxicity) when tested herein.

The filter disks and pads shall be wrapped (10 filter disks and 10 filter pads) to form an immediate package. Ten such immediate packages shall constitute one unit of issue. Immediate package shall have an indicator which shows when sterilization is complete. Each unit of issue and immediate package shall have instruction for interpreting the indicator.

Immediate package, disks, pads, and wrapping paper shall not deteriorate, buckle or deform and indicator shall perform as shown on instructions. Diameter of disk shall be 46.6 $\pm$ 0.6 mm after autoclaving. Disk shall be measured between 10 to 15 minutes after removal from the autoclave. Temperature and humidity for waiting period and measurement shall be 25 degrees Celsius (^{o}C) and 40% relative humidity.

Workmanship. The immediate package, filter disks and pads shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Package (PG). One package containing one hundred disks, as specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of S-I. The unit of product for examination purposes shall be one filter disk and pad.

Sampling for tests. Sampling for tests (dimensional) shall be conducted in accordance with MIL-STD-105, with an AQL of 0 (percent defective) and an inspection level of S-1. The unit of product for test purposes shall be one filter disk and pad.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to the following:

Porosity. Porosity shall be determined by one of the following methods:

a. Calculate the theoretical weight of a solid filter by multiplying the volume by the density of the cellulose ester polymer and determine the actual weight of the filter disk.

$$\frac{\text{Theoretical weight of a solid filter disk} - \text{Actual weight of filter}}{\text{Theoretical weight of a solid filter}} \times 100 = \text{percent porosity}$$

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b. Calculate the volume of the sample filter disk. Introduce mercury into the pores of the filter disk.

$$\frac{\text{Volume of mercury introduced}}{\text{Theoretical volume of filter}} \times 100 = \text{percent porosity}$$

Refractive index. The following test shall be performed:

Cut a 1 mm by 5 mm strip from a filter disk. Dimensions shall be approximate, and cutting shall be done with a sharp instrument so that smooth edges are obtained. Place the strip on a glass slide. Add a drop of oil to the strip and cover strip with a cover glass. Place the prepared slide on the stage of a microscope (magnification 100X) and position so that the microscope is focused on an edge of the strip (inter-face of the oil and strip). If the refractive indices of the oil and strip are different, a line called a Becke line will appear at the interface. Move the objective lens up or down so that the Becke line is moving from the strip to the oil. Upward movement of the lens indicates that the refractive index of the oil is greater than that of the strip. Downward movement of the lens indicates that the refractive index of the strip is greater than that of the oil. If the strip and Becke line cannot be brought into focus, the refractive index of the strip and the oil are the same. All oils used shall be within the tolerances specified herein, and each disk shall be tested with at least two oils, one of greater and one of lesser refractive index than the disk. (Theory of the refractive index test is given in the HANDBOOK OF CHEMICAL MICROSCOPY.)

Absorptivity. The following test shall be performed:

Place pad in a container which has a diameter slightly larger than that of the pad. Slowly add 1.7 ml of endo broth to the center of the pad and allow to stand for 10 minutes. If pad does not absorb all of the endo broth, repeat the above procedure with another pad using 1.3 ml of endo broth. If pad does absorb the entire 1.7 ml of endo broth, repeat the above procedure with another pad using 1.8 ml of endo broth.

Bubble point. The bubble point test shall be run in accordance with ASTM Designation F-316, using water as the test liquid.

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Diffusibility. Filter disk shall be dropped flat on water. Disk shall be observed for dry spots and diffusion through the disk shall be timed with a stopwatch. (Wet areas become darker while dry areas do not change in color intensity).

Water flow rates. Water flow shall be determined at 25°C at a pressure differential of 70 cm of mercury.

Organism retention. Test dilution shall be a phosphate-buffered or a saline suspension of 1.0×10^3 to 1×10^7 cells of a 24-hour culture of Serratia marcescens per ml. All equipment shall be sterile. Using forceps, a sterile filter disk that has been autoclaved twice shall be placed on the filter support base and the funnel unit shall be placed on the disk and secured by means of the appropriate attachment device. Test dilution (100 ml) shall be drawn through the test disk by applying a vacuum to the sterile flask. The disk shall then be rinsed three times with 10 ml of sterile dilution water. A sterile filter holder shall then be attached to another filtration flask and, using an aseptic technique, the filtrate from the test disk shall be passed through a reference disk. This reference disk shall be the subject disk from a lot which has been furnished to the Government and which meets the requirements of this specification; or reference disk shall be fabricated from the same material but shall have a smaller pore size than the subject item. After filtration, reference disk shall be placed on Wilson's Peptone Agar at 32°C for 24 hours. A zero count on the incubated referenced disk indicates sterility of the filtrate and retention by the test filter disk.

Toxicity. Test dilution shall be a phosphate-buffered suspension of 30 to 80 viable cells of a 24-hour culture of E coli per 0.1 ml. The addition of 0.1% peptone to the test dilution shall be acceptable.

Filter disk shall be placed on the filter base, and funnel unit shall be placed on the disk and made secure by an appropriate attachment device. Twenty ± 2 ml of sterile dilution water shall be added to funnel. 0.1 ml of E coli dilution shall be pipetted into the funnel. Dilution shall be drawn through the disk by means of a vacuum, and disks shall be placed on membrane (m) Plate Count Broth. With the same pipet, 0.1 ml of E coli dilution shall be delivered to the surface of 100 mm plate of Plate Count Agar. E coli dilution shall be spread over the entire surface of the plate with a sterile glass streaking rod so that the dilution is evenly distributed. Incubate the inverted disk and agar plates at 35°C ± 5° for 24 hours and count the resultant colonies.

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Five disks and five agar plates shall be run. The disks shall be considered non-toxic if the arithmetic mean of the count on the five disks is at least 90% of the arithmetic mean of the count on the five agar plates. The colonies on the disks shall measure at least 0.5 mm in diameter. If the disks do not meet the above criteria of 90% growth, toxicity testing procedures may be repeated with five more disks. If test is repeated, arithmetic mean shall be calculated by comparison with 10 (5 +5) disks, with 10 (5 + 5%) agar plates. If disks fail to meet the criteria of 90% growth on 10 disks, lot shall be rejected.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling and marking must be specified in the contract and/or order.

NOTES: The following National Stock Number is covered by this document:

NSN	Item Identification
6640-00-299-8692	DISK, FILTERING, MICROPOROUS, HYDROSOL TYPE, 47 mm Dia., 100s.

Endo Broth referred to under "TEST" is NSN 6550-00-687-8436 Endo Broth, Membrane Filter, 2 cc, 24s.

Wilson's Peptone Agar may be obtained from Baltimore Biological Laboratory.

The m Plate Count Broth is available from the Difco Company, Detroit, MI.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6640-1757

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	A-A-54752	21 SEPTEMBER 1992

3 DOCUMENT TITLE

DISK, FILTERING, MICROPOROUS

4 NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed*)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, [illegible])	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable) 7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete* '96.2

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